

Fourier Örnek ①

$$A \cos(7t) = ?$$

$$\cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2} \Rightarrow \frac{A}{2} e^{j7t} + \frac{A}{2} e^{-j7t}$$

$$\star \frac{-1}{j} = j = e^{j0.5\pi}$$

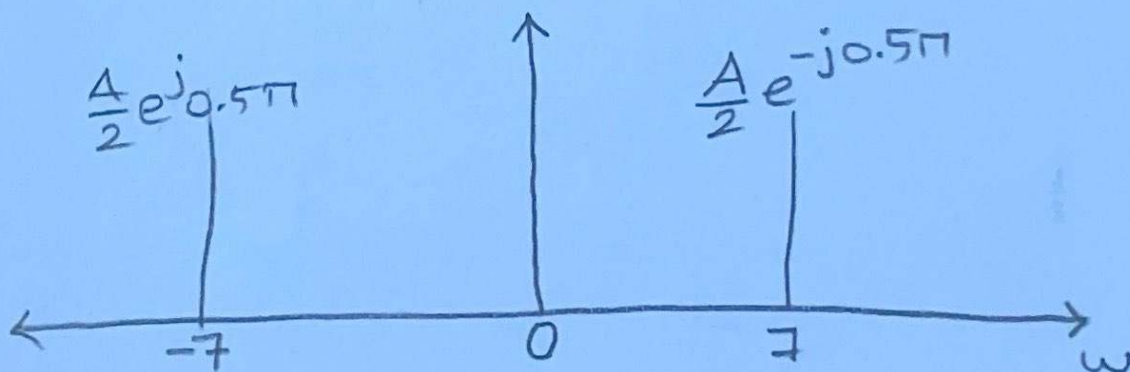
$$A \sin(7t) = ?$$

$$\frac{A}{2j} e^{j7t} - \frac{A}{2j} e^{-j7t}$$

$$\star \cos(\omega t + \phi) = \frac{e^{(j\omega t + j\phi)} + e^{(-j\omega t + j\phi)}}{2}$$

$$\frac{A}{2} e^{-j0.5\pi} e^{j7t} + \frac{A}{2} e^{j0.5\pi} e^{-j7t}$$

pozitif frekans fazı -0.5π
negatif " " 0.5π



Fourier Transform exercise 2

$$y'(t) + 2y(t) = x(t) + x'(t) \text{ freq. response}$$

$$h(\omega) = \frac{y(\omega)}{x(\omega)}$$

$$\begin{aligned} x(t) &\xrightarrow{F} x(\omega) \\ x'(t) &\rightarrow x(\omega)(j\omega)^1 \end{aligned}$$

$$(j\omega)y(\omega) + 2y(\omega) = x(\omega) + (j\omega)x(\omega)$$

$$\frac{y(\omega)}{x(\omega)} = \frac{j\omega + 1}{j\omega + 2} = h(\omega) = 1 - \frac{1}{2 + j\omega}$$

$$\underbrace{F^{-1}\{1\}}_{\delta(t)} - \underbrace{F^{-1}\left\{\frac{1}{2 + j\omega}\right\}}_{e^{-2t}u(t)} = h(t) = F^{-1}\{h(\omega)\} = F^{-1}\left\{1 - \frac{1}{2 + j\omega}\right\}$$

$$\delta(t) - e^{-2t}u(t) = h(t)$$

$$y(t) = h(t) * x(t)$$

Fourier örnek ③

$$x(t) = \cos(t) + 0.5 \cos(4t + \pi/3) + 0.25 \cos(8t + \pi/2)$$

Amp. formda ve $\omega_0 = 1$ iken exp. ve sine formadönüş-türünüz.

$$\begin{array}{lll} A_1 = 1 & A_4 = 0.5 & A_8 = 0.25 \\ \phi_1 = 0 & \phi_4 = \pi/3 & \phi_8 = \pi/2 \end{array} \quad \text{diğer } A_k \text{ ler } 0$$

$$\begin{array}{l} c_0 = A_0 \\ c_k = \frac{1}{2} A_k e^{j\phi_k} \\ c_{-k} = \frac{1}{2} A_k e^{-j\phi_k} \end{array}$$

$$c_1 = \frac{1}{2} \cdot 1 \cdot e^0 = 0.5 \quad c_4 = \frac{1}{2} \cdot \frac{1}{2} \cdot e^{j\pi/3}$$

$$c_{-1} = 0.5$$

$$c_{-4} = 0.25 e^{-j\pi/3}$$

$$c_8 = \frac{1}{4} \cdot \frac{1}{2} \cdot e^{j\pi/2} = 0.125 e^{j\pi/2} \quad c_{-8} = 0.125 e^{-j\pi/2}$$

$$x(t) = [0.5 e^{jt} + 0.5 e^{-jt}] + [0.25 e^{j\pi/3} e^{j4t} + 0.25 e^{-j\pi/3} e^{-j4t}] +$$

$$\downarrow \text{exp. form} \quad [0.125 e^{j\pi/2} e^{j8t} + 0.125 e^{-j\pi/2} e^{-j8t}]$$

$$\begin{array}{l} a_0 = c_0 \\ a_k = A_k \cos(\phi_k) \\ b_k = -A_k \sin(\phi_k) \end{array}$$

$$a_1 = 1 \cdot \cos(0) = 1 \quad a_4 = 0.5 \cos(\pi/3) = 0.25$$

$$b_1 = -1 \cdot \sin(0) = 0 \quad b_4 = -0.5 \sin(\pi/3) = -0.43$$

$$a_8 = 0.25 \cos(\pi/2) = 0 \quad b_8 = 0.25 \cdot \sin(\pi/2) = 0.25$$

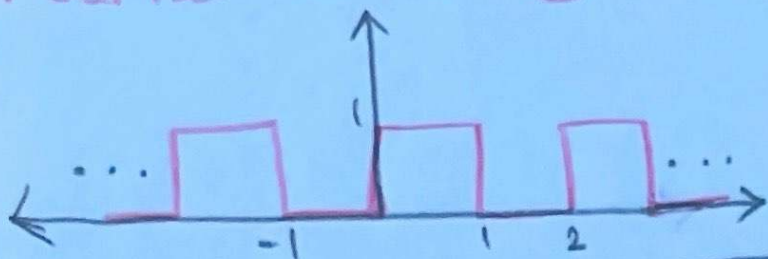
$$x(t) = \cos(t) + [0.25 \cos(4t) - 0.43 \sin(4t)] + [-0.25 \sin(8t)]$$

diğer a_k, b_k ler 0

$$\downarrow \text{sine-cosine form}$$

Fourier Örnek ④

FS coefficient bul.



sincosine form

① $T=2$

$\omega_0 = 2\pi/T = \pi$

$x(t) = \frac{1}{2} + \frac{2}{\pi} \sin(\omega_0 t) + \frac{2}{3\pi} \sin(3\omega_0 t) \dots$

$x(t) = \frac{1}{2} + \dots + \frac{-j}{-3\pi} e^{-j3\omega_0 t} + \frac{-j}{-1\pi} e^{-j1\omega_0 t} + \frac{1}{2} + \dots$
 $\frac{-j}{1\pi} e^{j1\omega_0 t} + \frac{-j}{3\pi} e^{j3\omega_0 t} + \dots$

② $C_k = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jk\omega_0 t} dt$
 $t_0 \leftarrow t_0 = 0 \text{ ol.}$

exp. form

③ $C_k = \frac{1}{2} \int_0^2 x(t) e^{-jk\pi t} dt = \frac{1}{2} \left[\int_0^1 1 e^{-jk\pi t} dt + \int_1^2 0 e^{-jk\pi t} dt \right]$

$= \frac{1}{2} \int_0^1 e^{-jk\pi t} dt = \frac{1}{2} \left[\frac{1}{-jk\pi} e^{-jk\pi t} \right]_0^1$

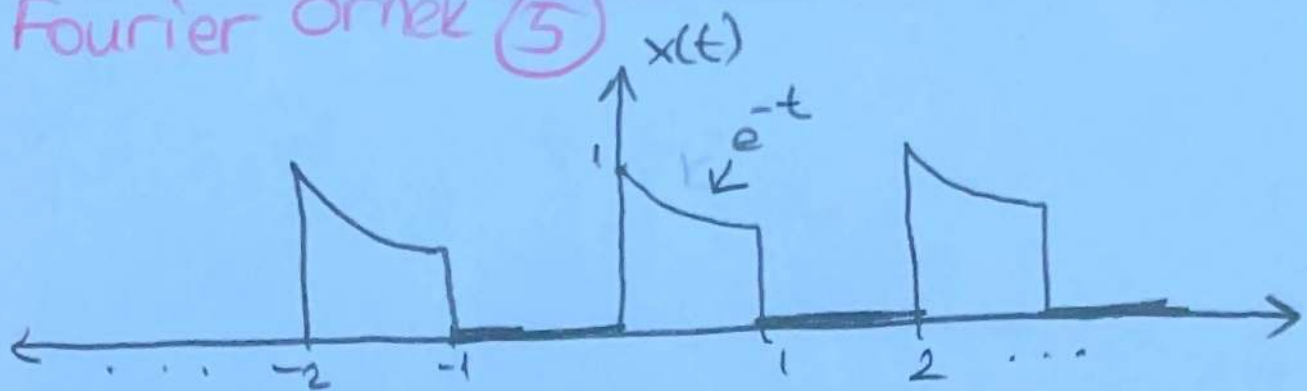
$\frac{-1}{j} = j$

$= \frac{j}{2k\pi} [e^{-jk\pi} - 1]$
 $\left. \begin{array}{l} e^{-jk\pi} = \cos k\pi - j \sin k\pi \\ k \text{ 'nin tek de\u011ferleri} \\ e^{-jk\pi} = -1 \end{array} \right\} \quad \left. \begin{array}{l} k \text{ 'nin a\u0131ft de\u011ferleri} \\ e^{-jk\pi} = 1 \end{array} \right\}$

$C_k = \begin{cases} 0, & k \text{ even, } \neq 0 \\ \frac{-j}{k\pi}, & k \text{ odd} \\ \frac{1}{2}, & k=0 \end{cases}$
 $C_0 = \frac{1}{2} \int_0^1 1 \cdot e^{-j0\pi t} dt = \frac{1}{2}$
 $a_0 = C_0$
 $\rightarrow a_k = 2 \operatorname{Re}\{C_k\}, k=1,2,\dots$
 $b_k = -2 \operatorname{Im}\{C_k\}$

$a_k = \begin{cases} \frac{1}{2}, & k=0 \\ 0, & k \neq 0 \end{cases} \quad b_k = \begin{cases} 0, & k \text{ even} \\ \frac{2}{k\pi}, & k \text{ odd} \end{cases}$

Fourier örnek ⑤



① $T=2$ $\omega_0 = \frac{2\pi}{T} = \pi$

② $c_k = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jk\omega_0 t} dt$
 $t_0 = 0$

$$e^{-jk\pi} = \begin{cases} 1, & \text{even } k \\ -1, & \text{odd } k \end{cases}$$

$$(-1)^k$$

$$= \frac{1}{2} \int_0^1 e^{-t} e^{-jk\pi t} dt + \int_1^2 0 dt$$

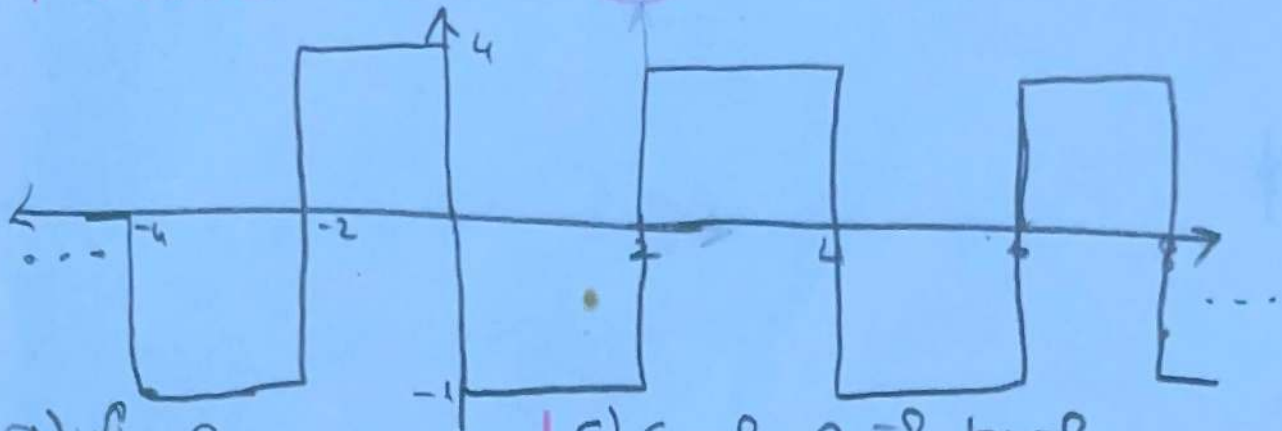
$$= \frac{1}{2} \int_0^1 e^{-t(1+jk\pi)} dt = \frac{1}{2} \left[\frac{e^{-t(1+jk\pi)}}{-1-jk\pi} \cdot -1 \right]_0^1 = \frac{-1}{2(1+jk\pi)} (e^{-(1+jk\pi)} - 1)$$

$$= \frac{1 - e^{-1} e^{jk\pi}}{2(1+jk\pi)} = \boxed{\frac{1 - e^{-1} (-1)^k}{2(1+jk\pi)}} = c_k$$

$$c_1 = \frac{1 + e^{-1}}{2(1+j\pi)}$$

$$c_2 = \frac{1 - e^{-1}}{2(1+2j\pi)} \dots$$

Fourier Örnek 6



a) $f_0 = ?$

$$T_0 = 4 \text{ ms} \quad f_0 = 1/T_0 = 0.25 \text{ Hz}$$

b) DC bileşen = ?

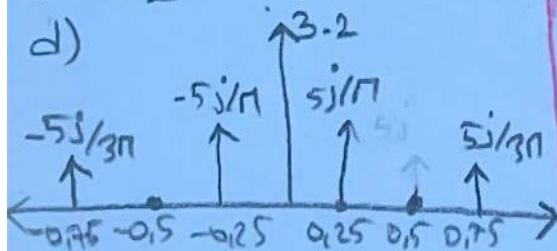
$$a_0 = c_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt$$

$$= \frac{1}{4} \left[\int_0^2 -1 dt + \int_2^4 4 dt \right]$$

$$= \frac{1}{4} \left[-t \right]_0^2 + \frac{1}{4} \left[4t \right]_2^4$$

$$= -1/2 + 4 - 2 = 1.5$$

d)



c) $c_k = ?$, $a_k = ?$, $b_k = ?$

$$c_k = \frac{1}{4} \left[\int_0^2 -1 e^{-jk\pi t/2} dt + \int_2^4 4 e^{-jk\pi t/2} dt \right]$$

$$= \frac{1}{4} \left[\frac{+1 \cdot e^{-jk\pi t/2}}{-jk\pi/2} \right]_0^2 + \frac{1}{4} \left[\frac{4 e^{-jk\pi t/2}}{-jk\pi/2} \right]_2^4$$

$$= \frac{-j}{2k\pi} \left[5e^{-jk\pi} - 4e^{-jk2\pi} - 1 \right]$$

$$= \frac{j(1 - 5(-1)^k + 4(-1)^{2k})}{2k\pi} = \frac{5j[1 - (-1)^k]}{2k\pi}$$

$$a_k = 2 \operatorname{Re}\{c_k\} = 0$$

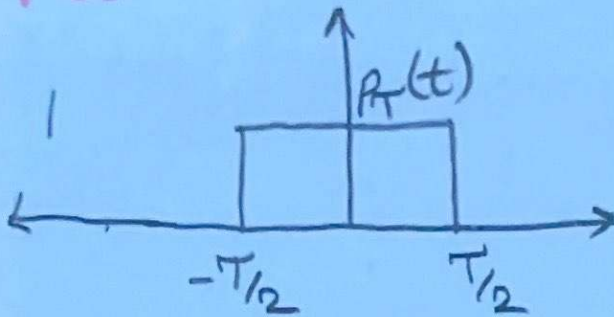
$$b_k = -2 \operatorname{Im}\{c_k\} = \frac{-5}{\pi k} (1 - (-1)^k)$$

$$c_1 = \frac{5j}{2\pi} \cdot 2 = \frac{5j}{\pi} \quad c_2 = 0 \quad c_{-2} = 0$$

$$c_{-1} = -\frac{5j}{\pi} \quad c_3 = \frac{5j}{3\pi} \quad c_{-3} = -\frac{5j}{3\pi}$$

Fourier Transform Örnek ①

find $P_T(\omega)$

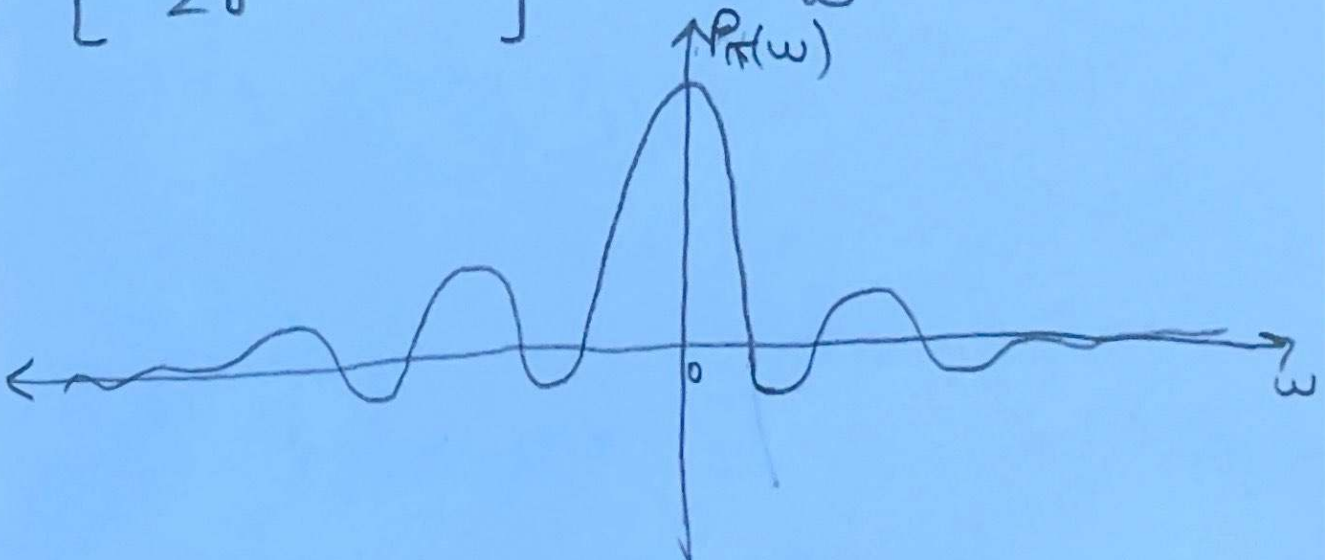


$$\textcircled{1} \quad P_T(t) = \begin{cases} 1 & (-\pi/2 \leq t \leq \pi/2) \\ 0 & \text{otherwise} \end{cases}$$

$$\textcircled{2} \quad P_T(\omega) = \int_{-\infty}^{\infty} P_T(t) e^{-j\omega t} dt = \int_{-\pi/2}^{\pi/2} 1 e^{-j\omega t} dt$$

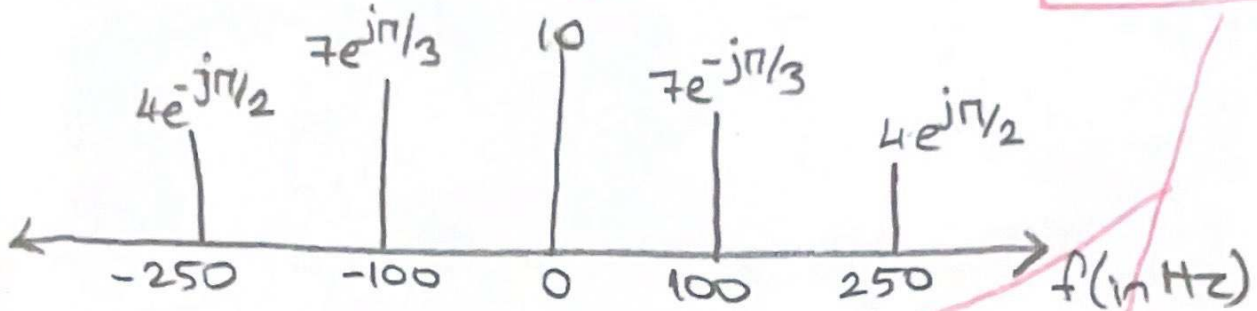
$$= \left[\frac{-1}{j\omega} e^{-j\omega t} \right]_{-\pi/2}^{\pi/2} = \frac{-1}{j\omega} \left[e^{-j\omega \pi/2} - e^{j\omega \pi/2} \right]$$

$$= \left[\frac{e^{j\omega \pi/2} - e^{-j\omega \pi/2}}{2j} \right] \frac{2}{\omega} = \frac{2 \sin(\omega \pi/2)}{\omega} = P_T(\omega)$$



Fourier Örnek (2)

$$\omega = 2\pi f t$$



$$X(t) = 10 + 7 \cdot e^{-j\pi/3} \cdot e^{j2\pi(100)t} + 7 \cdot e^{j\pi/3} \cdot e^{-j2\pi(100)t} + 4 \cdot e^{j\pi/2} \cdot e^{j2\pi(250)t} + 4 \cdot e^{-j\pi/2} \cdot e^{-j2\pi(250)t}$$

Orada tersine euler dönüşümü yaparsak

$$X(t) = 10 + 14 \cos(2\pi(100)t - \pi/3) + 8 \cos(2\pi(250)t + \pi/2)$$

frekans	faz	genlik
-250	$-\pi/2$	8
-100	$\pi/3$	14
0	0	10
100	$-\pi/3$	14
250	$\pi/2$	8

önce tablo
çizilir.

$$X(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi f_k t + \phi_k)$$

$$A_0 = 10$$

$$A_1 = 14$$

$$A_2 = 8$$

$$f_0 = 50$$

$$f_1 = 100$$

$$f_2 = 250$$

$$\phi_1 = -\pi/3$$

$$\phi_2 = \pi/2$$

Pratik ters euler

$$14 \rightarrow 14 \quad 4 \rightarrow 8$$

② frekansı poz. olanın fazı alınır. cos'un içine yazılır.

③ frek. poz. olan alınır.

Fourier Transform örneği ③

$$\frac{dy(t)}{dt} + 2y(t) = x(t) \quad y(t) = ?$$

$$a) x(t) = e^{-t} u(t) \quad b) x(t) = u(t)$$

$$(j\omega)y(\omega) + 2y(\omega) = x(\omega)$$

$$\frac{y(\omega)}{x(\omega)} = \frac{1}{2+j\omega} = h(\omega) \Rightarrow h(t) = e^{-2t} u(t)$$

$$a) x(\omega) = \frac{1}{1+j\omega} \quad y(\omega) = h(\omega) x(\omega) = \frac{1}{1+j\omega} \cdot \frac{1}{2+j\omega}$$

$$y(\omega) = \frac{1}{(1+j\omega)(2+j\omega)} = \frac{1}{1+j\omega} - \frac{1}{2+j\omega} \quad \left. \begin{array}{l} \text{ters} \\ \text{fourier} \end{array} \right\}$$

$$y(t) = e^{-t} u(t) - e^{-2t} u(t) = u(t) \cdot (e^{-t} - e^{-2t})$$

$$b) x(\omega) = \frac{1}{j\omega}$$

$$\frac{1}{j\omega} \cdot \frac{1}{2+j\omega} \Rightarrow \frac{1}{j\omega} \cdot \frac{1}{2+j\omega}$$

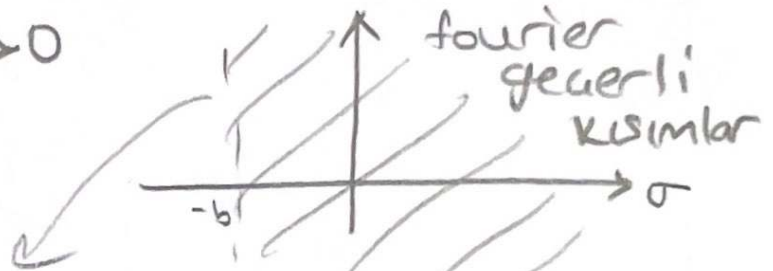
Laplace örneği ①

$$x(t) = e^{-bt} u(t).$$

$$(s+b > 0) \quad (b > 0)$$

$$X(s) = \int_0^{\infty} e^{-bt} e^{-st} dt = \int_0^{\infty} e^{-(s+b)t} dt = -\frac{e^{-(s+b)t}}{(s+b)} \Big|_0^{\infty}$$

$$= \frac{1}{s+b} \left\{ \begin{array}{l} \operatorname{Re}\{s\} + b > 0 \\ s + b > 0 \\ \sigma > -b \end{array} \right.$$



* $b > 0$ iken jw de bu alanda yani hem LT hem

FT var.

* $b < 0$ "

" bu alanda değil, yani sadece

LT var.

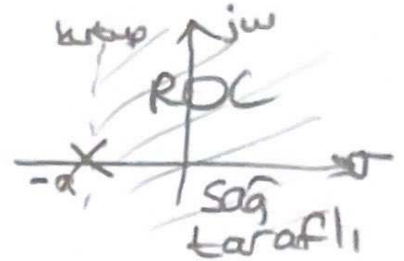
Laplace örneği (2)

$$x(t) = e^{-at} \cdot u(t)$$

$$x_2(t) = -e^{-at} u(-t)$$

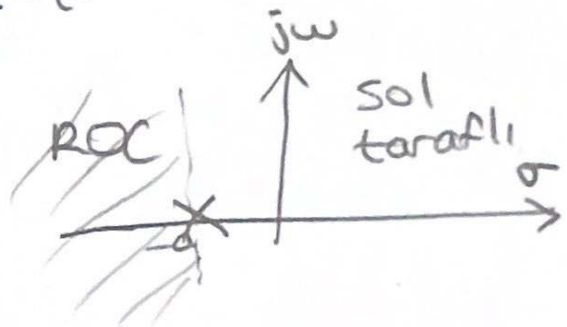
$$X(s) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-st} dt = \int_0^{\infty} e^{-(a+s)t} dt = e^{-(s+a)t} \bigg|_0^{\infty} \frac{-1}{(s+a)}$$

$$X(s) = \frac{1}{s+a} \quad (s+a > 0 \text{ ise}), (\sigma > -a)$$



$$X(s) = \int_{-\infty}^{\infty} -e^{-at} u(t) e^{-st} d(t) = \int_{-\infty}^0 -e^{-(a+s)t} dt$$

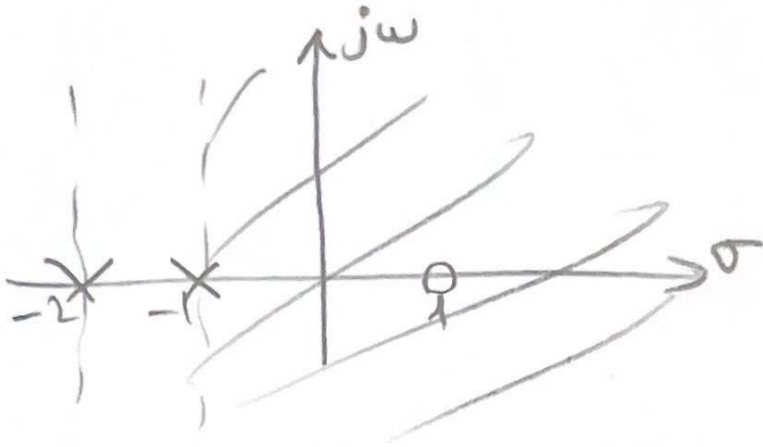
$$= + \left(\frac{e^{-(s+a)t}}{-(s+a)} \right) \bigg|_{-\infty}^0 = \frac{1}{s+a} \quad (s+a < 0 \text{ ise}), (\sigma < -a)$$



Laplace örnek (3)

$$x(t) = 3e^{-2t}u(t) - 2e^{-t}u(t)$$

$$X(s) = \frac{3}{s+2} - \frac{2}{s+1} \quad \begin{cases} \operatorname{Re}\{s\} > -2, \sigma > -2 \\ \operatorname{Re}\{s\} > -1, \sigma > -1 \end{cases}$$



$$X(s) = \frac{s-1}{(s+2)(s+1)} \quad \begin{cases} s-1=0 \\ s=1 \rightarrow \text{systemin zero'su} \end{cases}$$